

Chapter 1

The Problem and Its Scope

Introduction

Rationale of the study

Smartphone is an advanced mobile phone device which is designed to solve daily accessibility problems. It provides so many feature and allow more than make phone calls and send text messages. It has become a device highly in demand due to its power to perform basic and advanced computer functions. Any problem can be solved through one touch nowadays. That is why in modern life style, people can't live without it and they have become a necessity in life. The use of smartphone provides high quality performance and quick access to information and entertainment, such as mobile audio and video calls, mobile teleconferencing sending and receiving emails, and easy access to the internet for different kinds of people, including students. One more usage of it is entertainment social media. It is the source of all kinds of social connectivity and fun.

Smartphone addiction is the obsessive use of smartphones. As of 2024 there are over 6.8% billion smartphone users in the world and within that population 6.3% have a phone addiction problem. Research published by Virgin Mobile discovered that those billions of smartphone users receive 426% more messages and notifications than they did a

decade ago. They also send 278% more texts (2024 Addiction). Smartphone addiction is a real problem and cause of great concern. It impacts our health, relationships as well as work. People suffering from mobile addiction suffer from nomophobia which is fear of being without or unable to use your mobile phone for some reason or the other.

This issue of smartphone addiction and its impact on academic performance holds relevance at various levels, school environment. Schools are increasingly grappling with the challenge of managing smartphone usage among students and its impact on learning parental concern. Parents are worried about their children's excessive smartphone use and its potential consequences for their education and well-being. Local initiatives, local community may implement programs and initiatives to address smartphone addiction on academic performance is a significant concern with studies consistently revealing a negative correlation between the two. This impact can be attributed to distraction and reduced concentration, constant notification and the urge to check the phone destroy focus and make it difficult to concentrate on studies. Time management issues, sleep deprivation, and social isolation, spending excessive time on the phone can lead to reduced social interaction and isolation, impacting learning and development. Here in DUMANJUG NATIONAL HIGH SCHOOL smartphone addiction is also a common problem of the student. Male students are

addicted in playing games like mobile legends. Most of these students are in a higher level grade like grade 9, 10, 11 and 12. Some of the female students in the school are also addicted to smartphone because of social media like Facebook, Instagram, and TikTok

The prevalence rate of Smartphone addiction among Filipino adolescents is 62% (www.sciencedirect.com). Because of this high rate of smartphone addiction, it is a very common issue in our society. It also affects the mental and behaviour aspect of the teenagers. A recent study of smartphone addiction predictive capacity for health related quality of life among Filipino adolescents found a high prevalence rate of Smartphone addiction (62.6%) (Buctot, Kim, and Kian 2020). Ninety-nine percent of the 114-million population own a smartphone but only 7 in 10 Filipinos use the internet via mobile phones. Smartphone ownership and screen time are increasing across the world, but there have been few attempts to quantify Smartphone addiction on a global scale but it is common in few countries. (www.sciencedirect.com). 56% of Americans admit that they are addicted to their phones (explodingtopics.com) and it has been stated further that China tops the list for Smartphone addiction.

The purpose of this study is to delve into the complex relationship between smartphone addiction and academic performance among students. It aims to shed light on the factors that contribute to

smartphone addiction and to understand how this addiction impacts student academic success. Smartphone addiction is a complex issue with far reaching consequences, particularly for students. Understanding the determinants of this addiction and its influence on academic performance is crucial for developing effective interventions and promoting effective interventions and responsible digital citizenship.

STATEMENT OF THE PROBLEM

This study aims to investigate the determinants of smartphone addiction and its impact on academic performance among students at Dumanjug National High School

Specific Questions:

The study seeks to answer the following questions:

1. What is the level of smartphone addiction among the students of Dumanjug National High School?
2. What is the academic performance of the respondents in the first quarter?
3. What is the relationship between the level of smartphone addiction and academic performance?

4. What are the factors that contribute to smartphone addictions of the students?
5. What strategies or interventions can be recommended to address smartphone addiction and improve academic performance among students?

SCOPE AND DELIMITATION OF THE STUDY

This study aims to identify and analyze the determinants of smartphone addiction among students in Dumanjug National High School. This scope is limited to the students of Dumanjug National High School and the determinants of smartphone addiction includes: craving, social relationships (frequency and enjoyment of social communications), social relationships (behaviours and consequences), smartphone addiction (physical discomfort), smartphone addiction (consistency in work throughout the term).

This study will be limited to SHS students enrolled in Dumanjug National High School during the school year 2024-2025. It will not include students from other regions or educational levels. The study will focus on the overall impact of smartphone addiction on academic performance and will not delve into the influence of specific smartphone

apps. Additionally, the study will not explore other factors that may affect academic performance, such as socio-economic status, learning disabilities, or family support. The study will also not examine the long-term consequences of smartphone addiction or the effectiveness of potential interventions. Furthermore, the study will only include a sample of 242 students, selected through randomization due to time and resource constraints

RESEARCH HYPOTHESES

Null Hypothesis (Ho): There is no significant relationship between smartphone addiction and academic performance among students at Dumanjug National High School

Alternate Hypothesis (Ha): Smartphone addiction significantly influences the academic performance of the students at Dumanjug National High School.

BACKGROUND OF THE STUDY

Smartphones are the most wanted gadget in this era and smartphone addiction is associated with its use. (Bugti,2023). According to Nair et al, (2014), smartphones are electronic gadgets mostly used for two-way communication (Bugti,2023). Smartphone addiction is a disorder involving compulsive overuse of mobile devices and/or the total

amount of time they are online over a specified time. Compulsive smartphone use is just one type of technology addiction. However, other technologies prone to overuse like social media and gaming, are often accessed through mobile devices (Hanna & Wigmore,2024)

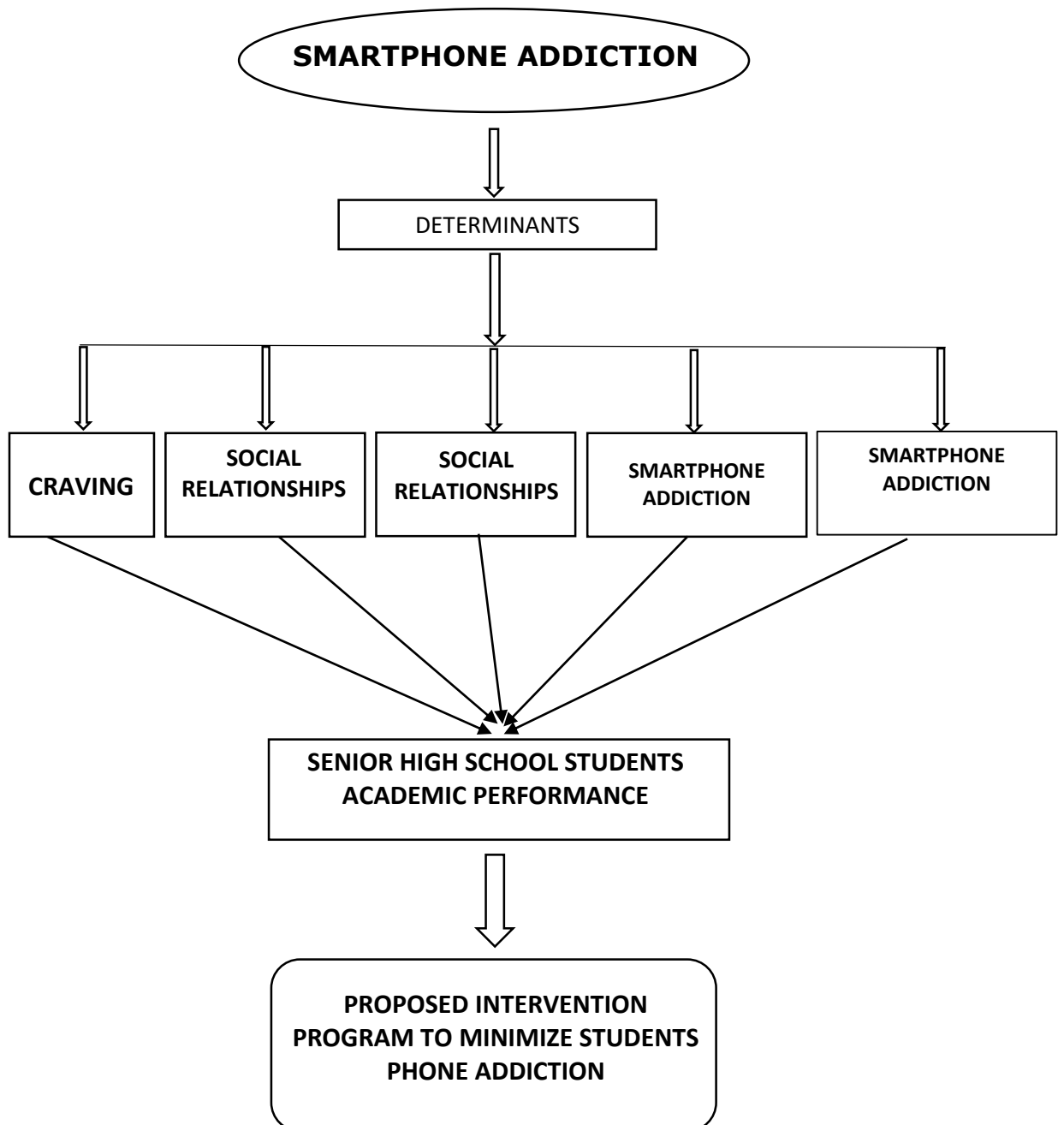
Boumsleh and Jaaloouk (2017) found an apparent association between smartphone use and academic achievement partly explained by the nature of the task the student is engaged in when using a smartphone. Smartphone addiction is increasingly affecting the masses and is negatively impacting the younger generation. However no work has been done to predetermine academic performance from smartphone addiction using data mining techniques (Chaudhury & Tripathy,2018)

There are few studies on smartphone addiction. In a study on smartphone addiction Cho and Kim (2014), identified gender average daily using time in a week, average daily using time in weekend, wrist pain in using smartphone, accident in using smartphone, sociality, impulsiveness, and social networking sites (SNS) addiction as significant predictors. They have found 43.3% explained variance of these factors in smartphone addiction.

While using smartphones people become unmindful, cause thousands of deaths and faulty activities. Its adverse effect is also seen in work-related tasks, classroom learning (Hiscock,2004;SELWYN,2003, and academic performance (Kuss & Griffiths,2011). In the classroom,

students engage in surfing the web, social networking, checking emails and text messages and consequently pay less attention to their lessons. Moreover, students spend more time with their smartphone that hampers their regular studies.

CONCEPTUAL FRAMEWORK OF THE STUDY



DEFINITION OF TERMS

Academic performance. Indicates a student's achievement after completing a course or subject from an institution. It measures student's learning across various academic subjects, which is assessed by formative and summative assessment.

Determinants. It is an element that determines the outcome.

Demographic Profile. It is the information of the respondents like gender, age, sex, etc.

Influence. The capacity to have an effect on the character, development, or behaviour of someone or something, or the effect itself.

Overuse of Smartphones. A long and uncontrollable usage of a phone for a long period of time even after trying to control one's usage of smartphone to get the same level of individual experience previously is referred to as overuse of smartphone.

Prevalence. It is the population of all the respondents.

Smartphones. A cellular telephone with an integrated computer and other features not originally associated with telephone.

Smartphone Addiction. A disorder involving compulsive overuse of mobile devices, usually quantified as the number of times users access

their devices and/or the total amount of time they are online over a specified period.

Student. Is a person who is a learner in the Dumanjug National High School.

CHAPTER 2

REVIEW OF RELATED LITERATURE

Today's youth are obtaining access to smartphones at younger and younger ages (Bennet,2020). Although there are many studies in the literature for the factors influencing the use of smartphone, the majority of these studies are for general use of mobile phones. The research on addition of mobile phone is limited (Zuhal and Volkan,2016).

Smartphone addiction, also referred to as problematic smartphone use, is a behavioural addiction characterised by excessive or compulsive use of smartphones. It has become a growing concern, particularly among students, due to it's potential to interfere with daily activities, including academic performance. This review synthesis key literature on the determinants of smartphone addiction and how it affects academic outcomes.

Smartphone is an electronic gadget mostly used for daily two-way communication (Nair, Vjay & Vinod,2014). It is a device used as a communication tool (Khan, Khalid & labal,2019). In the past few years, it has become so common that individuals, especially youth cannot even think of their lives without using smartphones (Philips & Bianchi,2005)

According to Sheila and Steve (2020) due to the high accessibility and mobility of smartphones, widespread and persuasive smartphone use has become the social norm, exposing users to various health and other risk factors. There is however a debate on whether addiction to smartphone is a valid behavioural addiction that is distinct from similar conditions. However according to Gath et al. (2024) smartphones are an unavoidable part of modern life. While some research suggests that they are incongruous with educational success and well-being for children and youth, other research advocates for the beneficial role it can play at school.

A smartphone is a gadget that can make and receive calls when connected to a network. This gadget does not require a person to stay in one place as they are able to move around within any geographical area that has cell network connectivity. A smartphone gets interconnected to an associated network provided by a mobile phone operator, allowing access to the public telephone network. Today, the smartphone is no longer limited to making and receiving calls. There are value-added services such as text messaging, sending email, online chatting, internet access, photography and gaming. In this borderless global era, mobile communication has become part of our life regardless of region, culture, age and community. Seong and Bo(2018) agreed that smartphone addiction is already entrenched with internet addiction as

the symptoms and effects, either positive or negative, are similar. Plant(2001) indicated that there is no standard smartphone effect although the same attitudes, rituals, responses and emotional engagements were discovered. Spending four hours on the smartphone can be considered as addiction, though such criteria of spending seven hours on the smartphone was commonly categorised as a sign of addiction.

Smartphone addiction could be considered a form of technological addiction (Derounian,2017). Nevertheless, apart from the person-to-person description, uncontrollable usage of smartphone can effect academic performance. Smartphone addiction can be categorized as a restlessness symptom associated with attempts to frequently check the phone even though there are no notifications (Ling,2024). Besides that, this addiction is defined as a strong desire to use the smartphone in order to remain occupied by surfing the internet, shopping online and playing games. Students interacting with their phones out of habit can be a way to avoid communicating with parents or teachers. Some students may experience withdrawal symptoms such as insomnia, anxiety, and depression when they are without their smartphone. Rice et al.(2018) exposed that smartphone usage relates to occupation, income and marital status of the potential users. Though many current studies have shown smartphone usage benefits in learning, Woodcock

et al. (2012) demonstrated contrasting findings. Students were found always using their phones more for playing games and other leisure activities than for learning. Among the identified factors preventing widespread adoption of smartphone for education are physical, technical and psychological limitations. White and Mills(2012) also found that students were increasingly adopting smartphones with the focus on personal use rather than education. In this study, students were asked if they used smartphone applications in categories in education, dictionary, games and maps. Similarly, Anzai et al.(2013) investigated mobile device use for mobile photo, note-taking, in order to examine the effect of this trend on vocabulary retention.

Determinants of Smartphone Addiction

Several psychological, social, and technological factors contribute to the development of smartphone addiction.

Psychological Determinants

- **Personality Traits:** Research has shown that certain personality traits are linked with higher risks of smartphone addiction. Cholz (2012) and Billieux et al. (2015) found that individuals with impulsive tendencies, low self-control, and emotional instability

are more likely to engage in compulsive smartphone use. Additionally, Elhai et al. (2017) showed that traits like neuroticism are associated with excessive smartphone use as a coping mechanism for emotional difficulties.

- **Anxiety & Stress:** Smartphone use as a stress-relief tool has been explored in multiple studies. Elhai et al. (2017) concluded that students with higher levels of anxiety and stress tend to use smartphones excessively, seeking comfort through social media and entertainment apps. Leung (2008) supported this, showing that stress and smartphone overuse form a vicious cycle, where stress leads to overuse, which in turn exacerbates stress.
- **Social Anxiety:** Hong et al. (2012) demonstrated that socially anxious individuals often turn to smartphone to escape real-life social interactions. The reliance on virtual interactions increases their likelihood of becoming addicted to smartphone, as they avoid in person communication.

CHAPTER 3

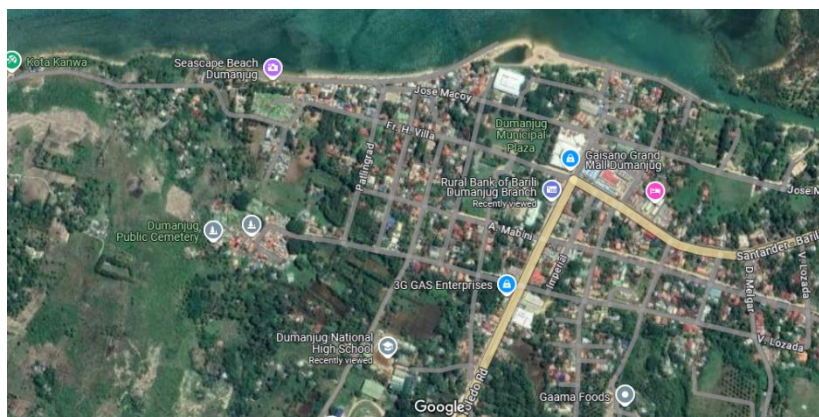
RESEARCH DESIGN AND METHODOLOGY

RESEARCH DESIGN

The study employed correlational research design because it compares two variables. The topic focuses on the determinants of smartphone addiction and its influence on academic performance of the students. The variables that are being compared is the smartphone addiction and students academic performance.

RESEARCH ENVIRONMENT

The study was conducted in Dumanjug National High School, one of the public schools of Dumanjug. It is located near Cebu Technological University (CTU) Dumanjug Extension in Li-ong Dumanjug Cebu. There are 62 teachers including school staffs and the principal, there are also approximately 2,000 students in this school. The schools mission is to facilitate, teach, assist, and give improvements to the students.



SAMPLE AND POPULATION

The sample size was calculated by using the Slovin's formula, $n = \frac{N}{1 + (0.5)^2}$. The formula is: $n = \frac{N}{1 + Ne^2}$, where: n is the sample size and N is the population size and e is the margin of error. The (n) stands for the whole population and 0.05 is the margin of error. So our population students size is 615, 340 students of grade 11 and 275 students of grade 12 students. So first get the 0.05 squared, so the outcome is 0.0025. Then multiply 0.0025 multiplied by 615 so the outcome is 1.5375, then you have to add 1 and 1.5375 equals 2.5375. Lastly divide 615 divided by 2.5375 equals 242.

RESEARCH INSTRUMENT

Our questionnaire is a Likert scale that is answerable by Strongly Disagree, Disagree, Neutral, Agree, Strongly Agree. We conducted surveys on Senior High School Students specifically the grade 11 to 12. The number of items in the questionnaire are as its stated 4 items about "CRAVING", 5 for "SMARTPHONE ADDICTION" and 4 for "SOCIAL RELATIONSHIP".

DATA COLLECTION

Data were collected through survey questionnaires answered by the SHS student in Dumanjug National High School face-to-face, ensuring direct engagement with each participant. This approach allows immediate clarification of any ambiguities in the questionnaire and fostered a higher response rate.

The questionnaires were carefully structured to explore various factors potentially influencing smartphone addiction among SHS students. Questions covered aspects such as daily usage patterns, preferred applications, perceived benefits and drawbacks of smartphone use, social influences, and the students' overall psychological well-being. We also included questions on demographic factors like age, gender, and academic performance to facilitate a comprehensive analysis of the collected data and identify potential correlations. The data was then interpreted and analyze as basis in formulating conclusion.

STATISTICAL TREATMENT OF THE DATA

To determine if there is no significant relation between determinants of smartphone addiction and academic performance of the Students we used Pearson's r correlation.

The table below indicates the data treatment for determinants of Smartphone Addiction.

RANGE	DESCRIPTION	INTERPRETATION
4.2 – 5.0	Strongly Agree	Strongly Addicted
3.4 – 4.1	Disagree	Moderately Addicted
2.6 – 3.3	Neutral	Neutral
1.8 – 2.5	Agree	Slightly Addicted
1.0 – 1.7	Strongly Disagree	Not Addicted

On the other hand, the academic performance will be interpreted below:

RANGE	INTERPRETATION
90 – 100	Outstanding
85 – 89	Very Satisfactory
80 – 84	Satisfactory
75 – 79	Fairly Satisfactory
Below 75	Not Addicted

Chapter 4

DATA PRESENTATION, ANALYSIS, AND INTERPRETATION

Table 1: The Determinants of Smartphone addiction of the Respondents.

Determinants	4.2 – 5.0 Strongly Addicted	3.4 – 4.1 Moderately Addicted	2.6 – 3.3 Neutral	1.8 – 2.5 Slightly Addicted	1.0 – 1.7 Not Addicted
Frequency	9	75	63	77	18
<u>CRAVINGS</u>	3.71%	30.99%	26.03%	31.81%	7.44%
%					
Frequency	23	40	90	65	24
<u>Social Relation- ship</u>					
%	9.51	16.33%	37.19%	26.85%	9.91%
Frequency	13	60	87	60	22
<u>Social Relation- ship</u>					
%	5.37%	24.79%	35.95%	24.79%	9.09%
Frequency	14	60	85	65	18
<u>Social Relation- ships</u>					
%	5.78%	24.79%	35.12%	26.85%	7.44%
Frequency	17	90	82	35	18
<u>Smartphone Addiction</u>					
%	7.02%	37.19%	33.89%	14.46	7.44%

The table presented the results concerning addiction levels across five categories: cravings, social relationships, smartphone addiction, social relationships, and smartphone addiction. Participants were categorized into five levels of addiction: strongly addicted, moderately addicted, neutral, slightly addicted, and not addicted. The frequency column likely represents the frequency or number of participants falling into each addiction level of each category. The percentage column shows the proportion of participants in each addiction level relative to the total number of participants in the category.

The provided data suggested that smartphone addiction has the highest frequency in moderately addicted. This is evident in the 37.19% for smartphone addiction, which represents the highest percentage within the group. This suggested that while these individuals may have a moderate level of smartphone addiction, their experience is characterized by a strong desire to use their phones, potentially leading to difficulty in controlling their usage.

These traits which were found in each component and were the reason for the given name were seemingly identified in other research. For instance, Hooper and Zou (2007) associated the use of smartphones

to six behaviour categories: habitual, dependent, compulsive, addictive, mandatory and voluntary so these authors suggest mobile phone use can be classified in these categories. Nonetheless, excessive use of mobile phones should not necessarily be seen as a negative addiction, as stated by Cassidy (2006), who points out that the person who spends a lot of time using this mobile technology could significantly reduce his/her chances of developing other harmful habits, such as alcohol and tobacco consumption.

TABLE 2: The Academic Performance of the Respondents

Academic Performance	90 – 100	85 – 89	80 – 84	75 – 79	Below 75
	Outstanding	Very Satisfactory	Satisfactory	Fairly Satisfactory	Did not meet expectation
frequency	156	65	16	5	
Percentage (%)	64.46%	26.86%	2.48%	2.07%	

In table 2, listed the percentage ranges defining each grade level, suggesting a standard grading system. The highest grade appears to be

90-100%, followed by decreasing ranges. The second row presents the frequency distribution, indicating the number of students within each grade range. For example, the number "5" likely represents five students scoring between 75% and 89%.

Students with outstanding performance in school typically exhibit a combination of several key characteristics that contribute to their success. They are commonly not addicted to smartphone or low percentage of smartphone addiction. And they also putting their study first before watching social media or playing games. Also some of them have no cellphones to use for academic purposes and they are dependent on their intellectual knowledge,

To fully interpret the table, additional information is needed. Specifically, translation of the non-English labels in the third row is crucial. This would provide context for the qualitative descriptions of each grade range. Without this translation, the analysis is limited to the quantitative data, which shows the distribution. It allows for a quick assessment of overall achievement and weakness within the class. However, the incomplete and ambiguous nature of the table, due to language and unclear column, limits its overall effectiveness (Kabir,2016).

Table 3: Summary of Students Smartphone Addiction

RANGE	FREQUENCY	PERCENTAGE	INTERPRETATION
4.2 – 5.0	3	1.23%	Strongly Addicted
3.4 – 4.1	44	18.18%	Moderately Addicted
2.6 – 3.3	135	55.79%	Neutral
1.8 – 2.5	52	21.48%	Slightly Addicted
1.0 – 1.7	8	3.30%	Not Addicted

Based on the data presented in the table, the majority of the respondents fall under the “Neutral” category, comprising 55.79% of the total. This indicates that a significant portion of the students have a moderate level of smartphone addiction. On the other hand, a smaller percentage of students are categorized as “Slightly Addicted” (21.48%), “Moderately Addicted” (18.18%), and “Strongly Addiction” (1.29%).

The findings suggested that a considerable number of students may have varying degrees of smartphone addiction, with a majority falling in the neutral range. This implies the need for interventions and strategies to address smartphone addiction among students to promote healthier digital habits and potentially improve academic performance.

It is said that according to Shiela and Steve (2020) due to high accessibility and mobility of smartphones, widespread and persuasive smartphone use has become the social norm, exposing users to various health and other risk factors. There is however a debate on whether addiction to smartphone is a valid behavioural addiction that distinct

from similar conditions. While some research suggest that they are incongruous with educational success and well-being for children youth, other research advocates for the beneficial role can play at school.

Table 4: The correlation analysis between Academic performance and Smartphone Addiction.

Weighted Mean	Academic Performance	Overall Mean
	89.9 (very satisfactory)	3.0 (Neutral)
Person's Correlation	0.0345	
P Value	0.59329	
Interpretation	No significant relationship	

The correlation analysis of the data reveals insights into the relationship between academic performance and weighted mean scores. With a Pearson correlation coefficient of 0.0345, there appears to be a weak positive correlation between Smartphone Addiction and Academic Performance. It also suggested that, on average, as the overall mean score in academics increases, there is a slight tendency for the weighted mean score to also increase. However, the correlation coefficient being

close to zero indicates that the relationship is minimal and may not be significant.

Furthermore, the calculated p-value of 0.59329 adds another dimension to the correlation analysis. The p-value represents the probability of observing a correlation as extreme as the one computed, under the assumption that there is no true correlation in the population. In this case, the relatively high p-value suggested that there is no significant correlation between academic performance and determinants of smartphone addiction.

In conclusion, while the correlation analysis shows a weak positive relationship between academic performance overall mean and weighted mean scores, the high p-value indicates that there is no correlation between smartphone addiction and academic performance of the students. It is important to delve deeper into the data and consider additional variables that could impact academic performance and weighted mean scores. Further investigation and a more comprehensive analysis are necessary to better understand the underlying factors influencing the relationship between these two metrics according to Ng et al,(2017).

CHAPTER 5

SUMMARY OF FINDINGS, CONCLUSION, AND RECOMMENDATIONS

Findings

This chapter is summarizing the key findings, drawing conclusions, and proposing actionable recommendations for improving the academic performance of the students of Dumanjug National High School.

The level of smartphone addiction among students at Dumanjug National High School appears to be significant, with a notable percentage of students as having moderate addiction. Surveys indicate that many students spend several hours daily on their smartphone often prioritizing social media and gaming over academic responsibilities. This addiction can lead to difficulties in managing time and other activities.

The academic performance of respondents in the first quarter shows that there are more outstanding (90-100) students than others, then following the very satisfactory which is (85-89), then Satisfactory which is (80-84), then Fairly satisfactory (75-79), fortunately there are no students that are in the below 75 grades. With the weighted mean of 89.9 which is very satisfactory.

There is no significant relationship between the level of smartphone addiction and the academic performance of the students. Because of the low level of smartphone addiction, they are not distracted in their studies and their academic performance. The main cause why they are not addicted to their smartphone is because they are limiting their time in using their smartphone.

There are many factors that contribute to smartphone addictions of the students. Risk factors for phone addiction include lack of awareness of education, personal predisposition, mental health conditions, perceived social pressure, lack of alternative coping mechanisms, boredom and escapism, availability and accessibility, and social influence. Therefore, people with low self-esteem, lack of confidence, isolation, and psychological disorders are more like to be addicted to the use of smartphone.

There are some strategies that can improve the academic performance of the students. These are Time Management, Active Learning Techniques, Setting Realistic Goals, Effective Study Habits, Regular Review and Revision, Seeking Help When Needed, Limiting Distractions, Balancing Study and Relaxation, and a Positive Mindset. By implementing these strategies and providing support and guidance to

students, educators can help enhance their academic performance and promote a positive learning experience.

Conclusion

Based on the findings from Chapter 4 of your research conducted at Dumanjug National High School, it has been determined that there is no significant relationship between smartphone addiction and academic performance among the students. Particularly noteworthy is the prevalence of smartphone addiction observed in Table 1, where the highest frequency was found to be in the moderately addicted category, accounting for 37.19% of the respondents. This suggests a considerable number of students may have a moderate level of smartphone addiction. This suggests a moderate level of smartphone addiction among students, highlighting the need for interventions to address this issue and promote healthier digital habits. Additionally, the correlation analysis between academic performance and smartphone addiction revealed a weak positive correlation, indicating a slight tendency for higher academic performance to be associated with lower levels of smartphone addiction. However, the high p-value suggests that there is no significant correlation between the two variables. Further investigation and consideration of additional factors are recommended to gain a deeper understanding of the relationship between smartphone addiction and academic performance among students as suggested by Shiela and Steve (2020) and Ng et al,(2017).

RECOMMENDATIONS

We can recommend several tips to help students avoid smartphone addiction:

1. Set limits: Establish specific time periods during the day when you use your phone, and stick to those limits.

2. Create a schedule: Plan your day in advance and allocate time for studying, socializing, exercising, and other activities that don't involve your phone.

3. Use app blockers: Consider using apps or features that limit your access to certain apps or websites that may be contributing to your smartphone addiction.

4. Practice mindfulness: Be aware of your phone usage and how it affects your mood, productivity, and relationships. Take breaks and practice mindfulness techniques to stay present and focused.

5. Find alternative activities: Replace phone usage with other activities you enjoy, such as reading a book, going for a walk, or having a conversation with a friend.

6. Engage in hobbies: Cultivate hobbies and interests that don't involve your phone, such as art, music, sports, or cooking.

7. Socialize offline: Make an effort to spend time with friends and family in person, rather than relying solely on digital communication.

8. Turn off notifications: Disable unnecessary notifications on your phone to reduce the temptation to constantly check it.

9. Designate phone-free zones: Establish certain areas in your home or specific times of day where phone usage is not allowed, such as during meals or before bedtime.

10. Practice digital detox: Take regular breaks from your phone, such as setting aside a "tech-free" day once a week to disconnect and focus on other activities.

11. Use a physical timer: Set a timer when using your phone for a specific purpose or leisure activity to help you stay mindful of your screen time.

12. Seek support: Talk to friends, family, or a counselor about your struggles with smartphone addiction and ask for their understanding and encouragement as you work towards reducing your phone usage.

13. Reflect on triggers: Identify the situations or emotions that lead you to use your phone excessively and develop healthy coping strategies to address them, such as going for a walk, practicing deep breathing, or journaling.

Remember, it's essential to be patient and kind to yourself as you navigate the process of reducing your smartphone usage. Small changes and consistent efforts can make a significant difference in your life.

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APPENDICES

Appendix A

Research Output

Interventions

Goal- To encourage students to use their smartphones more sparingly, we should implement more controlled manners of owning smartphones and convince the parents to let us not abuse our online presence since it may be dangerous.

Objectives:

- Time regulations for mobile devices during class and break times for a more controlled smartphone usage habit.
- An age limit to allow the usage of smartphones especially the younger students and children so they become more aware that using smartphones instead of studying can be detrimental
- Insure that smartphone addiction will decrease after following disciplinary measures in turn reducing the likelihood of addiction.

Actions:

1. Set limits:

- A. Have a timeframe and use it within that time frame ex. Having a 2 hour time limit for morning, afternoon and evening
- B. Only use your smartphone when the situation calls for it, for example an emergency, and taking down notes.

2. Create a schedule:

- A. Use your phone during certain times per day like 2 hours in the morning and 2 hours in the evening so you can focus on the more important things in life.
- B. Be disciplinary on your child's screen time per day like sitting a limit for a certain time of the day.

3. Use app blockers:

- A. Only use these for children below the age of 18 so they don't accidentally look something they did not intend to look up such as online pornography and website with shady activities
- B. If you insist, put one on your smartphone and so you don't accidentally open a specific app that you still need to be installed for work or professional purposes.

Timelines:

- From infancy to the age of 13, children shouldn't have mobile devices with free access to the internet, it should only be with contacts for family members and friends, for parent approved video games, and educational resources with a time limit for leisurely activities.
- From ages 13 to 15, some restrictions should be lifted or relieved like extending the timeframe from 30 minutes to 1 to 3 hours and allowing them to use search engines with safe search on with parental controls
- From ages 15 to 18/21, more restrictions should be lifted due to being close to becoming young adults and since some of the topics or future projects can require the safe search engine to be turned off and the timeframe should be close to none since they are now more responsible with the resources they have in their fingertips.

Expected outcomes:

- A decrease of recorded smartphone addicts and an increase of responsible smartphone using students.
- A lot less fights happened due to being rather respectful in the internet resulting with better relationships

- Improved mental health due to less dependence on smartphones for online attention.
- Better academic performance because the usage of smartphones is to help with school such as research and other subjects.

Appendix B

Letter for the Principal

OCTOBER 17,2024

RICARDO I. FERRER, JR.

PRINCIPAL 1

DUMANJUG NATIONAL HIGH SCHOOL

LI-ONG, DUMANJUG, CEBU

Dear Sir:

GREETINGS OF PEACE

In partial fulfillment of our requirements for our subject Practical Research 2, we are conducting a research study entitled "Determinants of Smartphone Addiction and Its Influence on Academic Performance of the Students".

In connection with this, we would like to ask permission from your good office to allow us to administer our research questionnaire to the Senior High School Students of Dumanjug National High School. This will be conducted during the first semester of School Year 2024-2025.

Rest assured that all the data gathered will remain absolutely confidential and to be used in academic purposes only. We believe that you are one with us in our enthusiasm to finish the requirements as compliance for our subject and to develop our well-being. We hope for your positive response on this humble matter. Your approval to conduct this study will be greatly appreciated.

Respectfully Yours,

JE-IN MALIG-ON

ROMEO CATADMAN

WINTER DENIEGA

JHON PAUL BAONIL

FRANCIS BELACHO

WARREN GARSULA

Recommending Approval:

GERALDINE L. PORFERIO

Practical Researcher 2 Teacher

Approved by:

RICARDO I. FERRER, JR.

School Principal

Appendix C**Letters for the Students**

October 17,2024

Dear Respondents,

We are the Technical Vocational Livelihood major in Computer System Servicing (CSS) in Dumanjug National High School who are enrolled as Grade 12 Students. Presently we are conducting a study entitled "Determinants of Smartphone Addiction and its influence on academic performance of the students".

In this regard, we are asking for your precious time, and effort to answer all the questions in our research questionnaire that are important in the completion of our research.

Rest Assured that all data gathered from you will be kept in the highest level of confidentiality.

Your Positive response in this request will be a valuable contribution for the success of the study and will be highly appreciated.

Respectfully yours,

JE-IN MALIG-ON ROMEO CATADMAN WINTER DENIEGA
JHON PAUL BAONIL FRANCIS BELACHO

Noted by:

GERALDINE L. PORFERIO

Research Teacher

Appendix D

Questionnaire

Determinants of Smartphone Addiction Questionnaire

Adopted from <https://files.eric.ed.gov/fulltext/EJ1413924.pdf>

Name (Optional): _____ Section: _____

First Quarter GPA: _____ Year Level: _____

Directions: Write check (✓) in the box if your strongly agree, disagree, neutral, agree, or strongly agree about the question provided.

Prompts: CRAVING	(1) STRONGLY DISAGREE	(2) DISAGREE	(3) NEUTRAL	(4) AGREE	(5) STRONGLY AGREE
• I feel anxious when I forgot to take my smartphone with me					
• I feel anxious when I am not able to use my smartphone					
• I feel impatient and restless when my smartphone is unavailable					
• I feel anxious when I have not prepared my charging pack or battery for my smartphone					
• It would be difficult if I was not allowed to use a smartphone					

Prompts: SOCIAL RELATIONSHIP (frequency and enjoyment of social communications via smartphone)	(1) STRONGLY DISAGREE	(2) DISAGREE	(3) NEUTRAL	(4) AGREE	(5) STRONGLY AGREE
My social communications via smartphone are more frequent than in real life					
My social communications via smartphone are more enjoyable than in real life					
My social relationships with smartphone friends are more intimate than with my real life friends					
I express my true feelings better while communicating through smartphone rather than talking face-to-face					

Prompts: SOCIAL RELATIONSHIP (behaviors and consequences)	(1) STRONGLY DISAGREE	(2) DISAGREE	(3) NEUTRAL	(4) AGREE	(5) STRONGLY AGREE
I feel pain in the wrists or at the back of the neck while using a smartphone					
I have my smartphone in my mind even when I not using it					

The people around me tell me that I use my smartphone too much					
I don't really want to go to places where smartphone signals are weak					

Prompts: SMARTPHONE ADDICTION (physical discomfort while using smartphone)	(1) STRONGLY DISAGREE	(2) DISAGREE	(3) NEUTRAL	(4) AGREE	(5) STRONGLY AGREE
I have missed planned work due to smartphone use					
I use my smartphone for longer periods than I had intended					
I have tried to shorten my smartphone use time, but always fail					
Spending a lot of time on my smartphone has become a habit					
I need to spend an increasing amount of time using my smartphone to achieve the same satisfaction as before					
I involuntarily touch my smartphone					

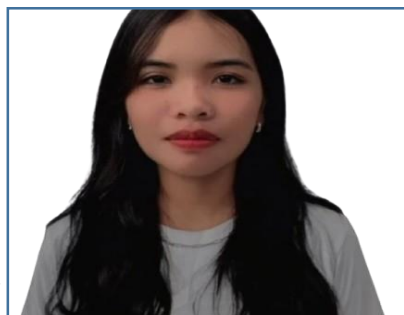
Prompts: SMARTPHONE ADDICTION (study habits and academic achievement)	(1) STRONGLY DISAGREE	(2) DISAGREE	(3) NEUTRAL	(4) AGREE	(5) STRONGLY AGREE
I work consistently throughout the term and review regularly when the exams are close					
I do all of my assignments as soon as possible after they have been set					
While studying I often think of real life situations to which the material that I am learning would be useful					
I usually become increasingly absorbed in my work the more I do					
I find that at times studying gives me a feeling of deep personal satisfaction					

Curriculum Vitae

JE-EN M. MALIG-ON

E-Mail Address: jeinmaligon7@gmail.com

Mobile: 09671798178

**PERSONAL INFORMATION**

Date of Birth: September 16, 2006

Civil Status: Single

Gender: Female

Religion: Roman Catholic

Citizenship: Filipino

Age: 18

EDUCATIONAL BACKGROUND

Elementary: Budlasan Elementary School

Trinidad, Guihulngan, City

Secondary: Dumanjug National High School

Li-ong, Dumanjug, Cebu

JHON PAUL A. BAONIL

E-Mail Address: paulbaonil@gmail.com

Mobile: 09859032274

**PERSONAL INFORMATION**

Date of Birth: September 13, 2007

Civil Status: Single

Gender: Male

Religion: Roman Catholic

Citizenship: Filipino

Age: 17

EDUCATIONAL BACKGROUND

Elementary: Dumanjug Central Elementary School

Li-ong, Dumanjug, Cebu

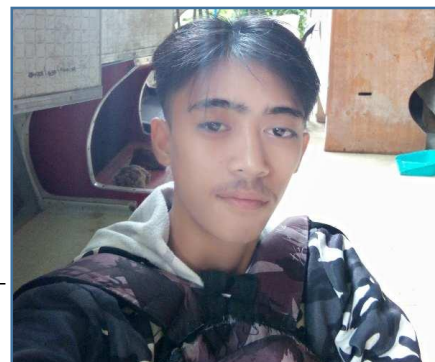
Secondary: Dumanjug National High School

Li-ong, Dumanjug, Cebu

ROMEO CATADMAN

E-Mail Address: catadmanromeo@gmail.com

Mobile: 09945773403

**PERSONAL INFORMATION**

Date of Birth: October 5, 2006

Civil Status: Single

Gender: Male

Religion: Catholic

Citizenship: Filipino

Age: 18

EDUCATIONAL BACKGROUND

Elementary: Lapaz Elementary School

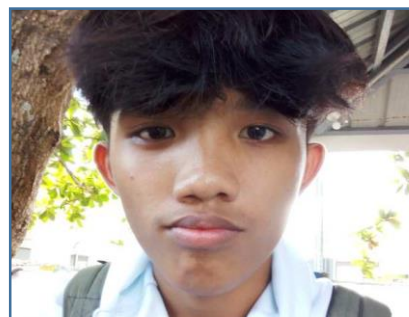
Secondary: Dumanjug National High School

Li-ong, Dumanjug, Cebu

FRANCIS F. BELACHO

E-Mail Address: francisbealcho3@gmail.com

Mobile: 09918281371

**PERSONAL INFORMATION**

Date of Birth: April 20,2007

Civil Status: Single

Gender: Male

Religion: Roman Catholic

Citizenship: Filipino

Age:17

EDUCATIONAL ATTAINMENT

Elementary: Sta. Cruz Elementary School

Sta. Cruz, Ronda, Cebu

Secondary: Dumanjug National High School

Li-ong, Dumanjug, Cebu

WINTER DENIEGA

E-Mail Address: winterdeniega0@gmail.com

Mobile: 09662568626

**PERSONAL INFORMATION**

Date of Birth: February 24, 2007

Civil Status: Single

Gender: Male

Religion: Roman Catholic

Citizenship: Filipino

Age: 17

EDUCATIONAL BACKGROUND

Elementary: Yati Elementary School

Yati, Liloan, Cebu

Secondary: Dumanjug National High School

Li-ong, Dumanjug, Cebu